

Labwork 3.2 - Pattern Drawing Machine

General Information and Submission Rules:

- In this labwork, you need to design a machine that prints different patterns based on the user choice.
- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork3.c**.
- Incorrect naming or file format will result in a loss of 5 points from your total grade.
- Only use concepts covered in the labs so far.

Question 1 - Display Message (15p)

- Create a new function (**welcome**) that displays a welcome message and a list of all available operations to the user.
- This function will expect a decision from a user so that it will execute related functions based on this choice until the user exits.
- Make sure to print each line separately.

```
Welcome to the Pattern Drawing Machine! By using this machine you can do:
1. Draw Palindrome Shapes
2. Check Palindrome Shape (6 input)
3. Draw a point on XY plane
4. Display Pascal Triangle
5. Exit
Make your choice:
```

Question 2 - Draw Palindrome Shapes (25p) and Check Palindrome Shape (15p)

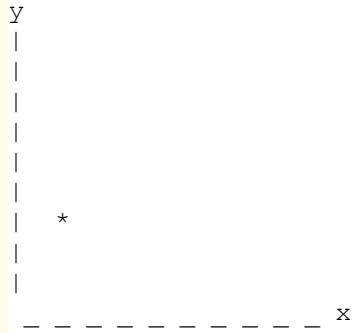
- Create new functions (**draw_palindrome** and **check_palindrome**). The explanations are given below:
- A **palindrome** is a word, phase, or a sequence of characters which reads the same backward as forward. For example, the word “madam” is a palindrome word. In this example we apply this definition to shapes.
- Write a **draw_palindrome** function that will create a palindrome triangle. It will take the (*base*) from the user. Ensure base is an odd number for symmetry. The horizontal line needs to be created with **asterisks (*)**. And other parts will repeat the **@** and **+** characters. Example is given below:
- In the **check_palindrome** function, it will expect a 6 character from the user and check if the characters follow palindrome rules. If not, return an error message to the user and ask shape until he/she gives a palindrome shape.

```
Make your choice: 1
Enter the base: 7
      *
    @*@
  @+*+@
@+@*@+@
Make your choice: 2
Enter the shape: *%+*%+
It is not a palindrome shape!
Enter the shape: *%++%*
It is a palindrome shape!
Make your choice:
```

Question 3 - Draw Point (20p)

- Create a new function (**draw_point**) that displays a point in the coordinate system (only Quadrant I need to be displayed) by using **asterisk (*)**.
- It will take the x and y coordinate from the user and show it on the plane. Assume that the maximum of x and y is 9.
- On the x line, there is space between two underscores ().

```
Make your choice: 3
Enter x and y coordinates: 2 3
```



Question 4 - Display a Pascal Triangle (25p)

- Create a new function (**pascal_triangle**) that displays pascal triangle shape by using asterisks (*).
- **Pascal's Triangle** is a mathematical order consisting of numbers arranged in a triangular pattern, which plays an important role in combinatorics. Each number is the sum of the two numbers immediately preceding it.
- Take the base of the shape from the use (base). Ensure base is an even number for symmetry. Assume the base is power of 2.
- If the shape cannot be drawn, print an error to the user and then ask the base again.

Hint:

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1
```

Enter the choice: 4

Enter the base: 9

It cannot be drawn since the base is odd!

Enter the base: 8

```
  *
  * *
 *  *
* * * *
 *      *
* *      * *
* *      * *
* *      * *
* *      * *
* *      * *
* *      * *
* *      * *
```

Enter the choice: 5

Good bye!